Lab 3 - Routing Protocol

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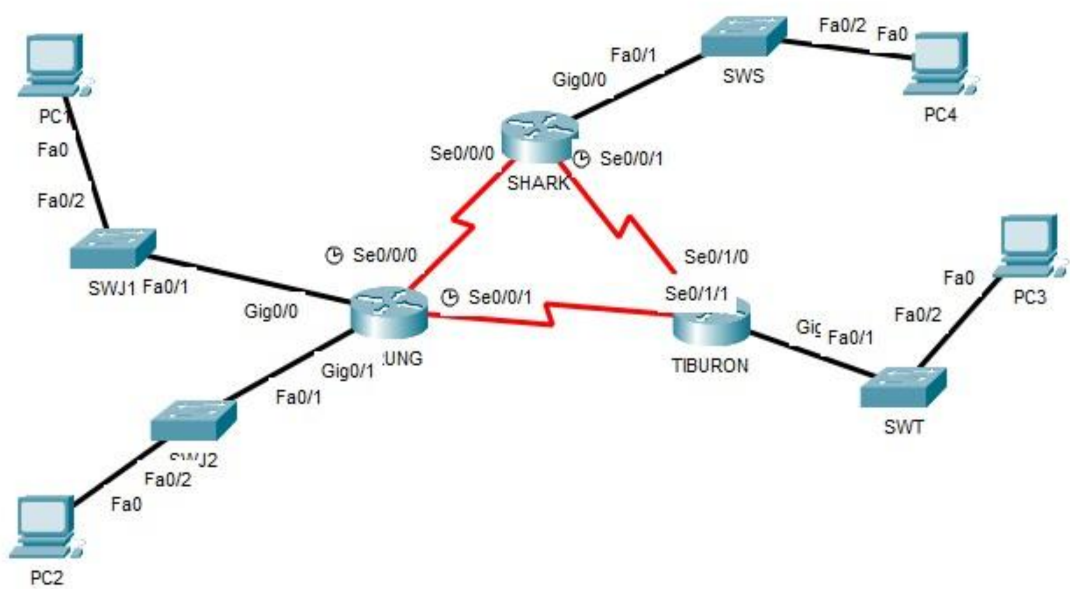
SECTION: **02**

INTRODUCTION

This lab looks into the network layer, focusing on subnetting and routing.

TOPOLOGY



LAB INFORMATION

Here is some basic information for the lab. The network address given is 172.18.110.0/23.

Table 1

Device	Subnetwork	Usable Hosts
JERUNG	LAN1	20
	LAN2	30
SHARK	LAN	60
TIBURON	LAN	50
Connections	JERUNG-SHARK	2
	JERUNG-TIBURON	2
	SHARK-TIBURON	2

Table 2

#	Device Name	Interface	IP Address	Subnet Mask	Gateway
1	JERUNG	Se0/0/0	172.18.110.193	255.255.255.252	
2		Se0/0/1	172.18.110.197	255.255.255.252	
3		G0/0	**172.18.110.129**	**255.255.255.224**	
4		G0/1	**172.18.110.161**	**255.255.255.224**	
5	TIBURON	Se0/1/1	**172.18.110.198**	**255.255.255.252**	
6		Se0/1/0	**172.18.110.202**	**255.255.255.252**	
7		G0/0	**172.18.110.65**	**255.255.255.192**	
8	SHARK	Se0/0/0	**172.18.110.194**	**255.255.255.252**	
9		Se0/0/1	172.18.110.201	255.255.255.252	
10		G0/0	**172.18.110.1**	**255.255.255.192**	
11	PC1	-	**172.18.110.158**	**255.255.255.224**	**172.18.110.129**
12	PC2	-	**172.18.110.190**	**255.255.255.224**	**172.18.110.161**
13	PC3	-	**172.18.110.126**	**255.255.255.192**	**172.18.110.65**
14	PC4	-	**172.18.110.62**	**255.255.255.192**	**172.18.110.1**

LAB TASKS

Task 1 – IP Addressing

1. Given the network address of the organisation and the basic information provided in both Tables 1 and 2. Show your workings here and complete Table 2 with the correct information. PCs will be given the last usable address of the subnetwork.

Jerung Lan 1 $2^x > 22$, $x = 5$

Lan 2 $2^x > 32$, $x = 5$

Shark Lan $2^x > 62$, $x = 6$

Tiburon Lan $2^x > 52$, $x = 6$

Jerung - Shark $2^x = 4$, $x = 2$

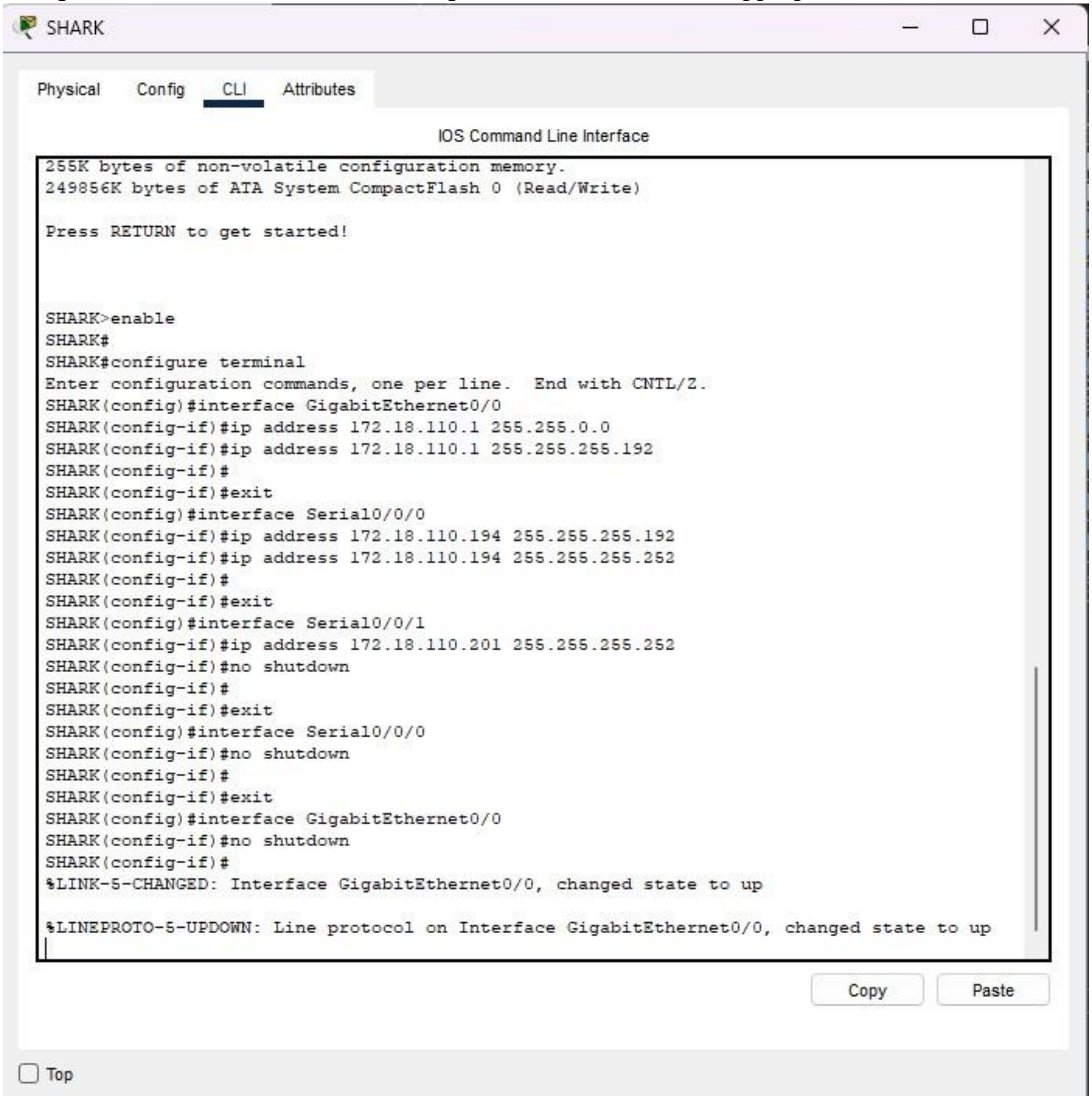
Jerung - Tiburon $2^x = 4$, $x = 2$

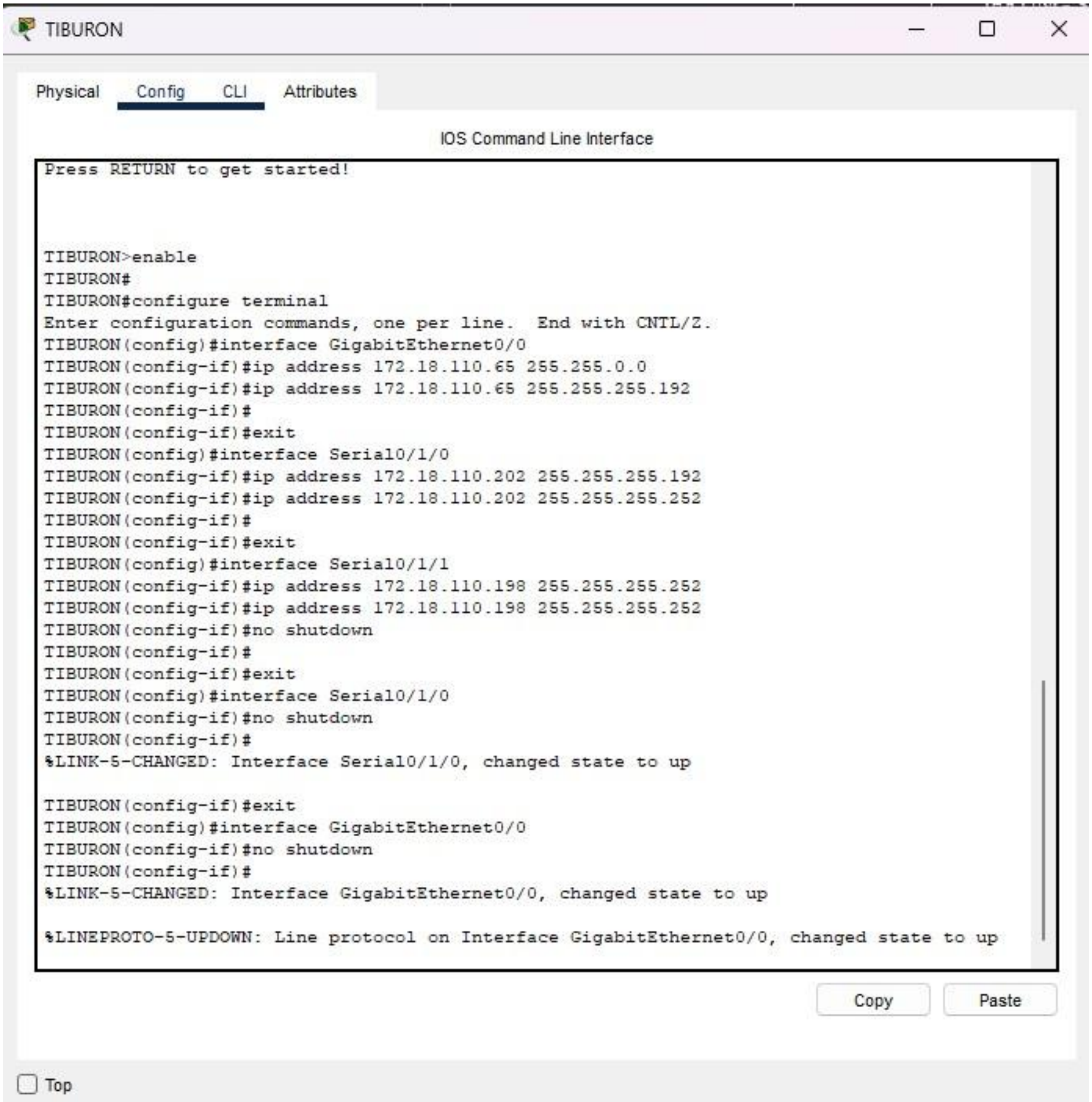
Shark – Tiburon $2^x = 4$, $x = 2$

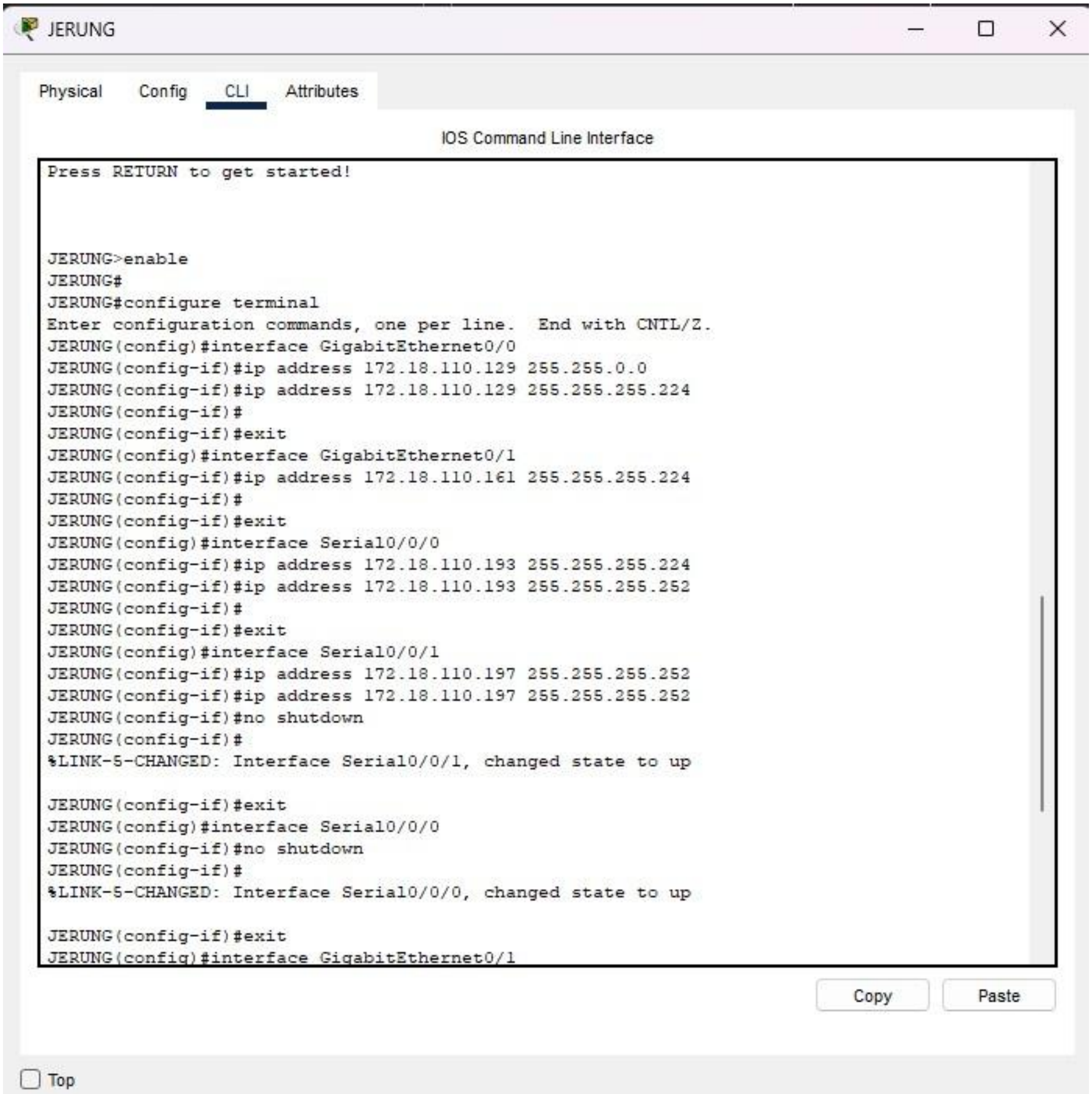
172.18.0110 111	0 . 0000 0000 [0]
S_0 - Shark	0. 0011 1111 [63]
S_1 - Tiburon	0. 0100 0000 [64]
	0. 0111 1111 [127]
S_2 - Jerung LAN 1	0. 1000 0000 [128]
	0. 1001 1111 [159]
S_3 - Jerung LAN 2	0. 1010 0000 [160]
	0. 1011 1111 [191]
S_4 - Jerung-Shark	0. 1100 0000 [192]
	0. 1100 0011 [195]
S_5 - Jerung - Tiburan	0. 1100 0100 [196]
	0. 1100 0111 [199]
S_6 - Shark - Tiburan	0. 1100 1000 [200]
	0. 1100 1011 [203]

Subnet	Network Address	Broadcast Address	Useable Address Range	Subnet Mask
0	172.18.110.0	172.18.110.63	172.18.110.1 – 172.18.110.62	255.255.255.192
1	172.18.110.64	172.18.110.127	172.18.110.65 – 172.18.110.126	255.255.255.192
2	172.18.110.128	172.18.110.159	172.18.110.129 – 172.118.110.158	255.255.255.224
3	172.18.110.160	172.18.110.191	172.18.110.161 – 172.18.110.190	255.255.255.224
4	172.18.110.192	172.18.110.195	172.18.110.193 – 172.18.110.194	255.255.255.252
5	172.18.110.196	172.18.110.199	172.18.110.197 – 172.18.110.198	255.255.255.252
6	172.18.110.200	172.18.110.203	172.18.110.201 – 172.18.110.202	255.255.255.252

2. Using the IP addresses calculated, configure the devices with the appropriate information.




TIBURON
Physical
Config
CLI
Attributes
IOS Command Line Interface
Press RETURN to get started!
TIBURON>enable
TIBURON#
TIBURON#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
TIBURON(config)#interface GigabitEthernet0/0
TIBURON(config-if)#ip address 172.18.110.65 255.255.0.0
TIBURON(config-if)#ip address 172.18.110.65 255.255.255.192
TIBURON(config-if)#
TIBURON(config-if)#exit
TIBURON(config)#interface Serial0/1/0
TIBURON(config-if)#ip address 172.18.110.202 255.255.255.192
TIBURON(config-if)#ip address 172.18.110.202 255.255.255.252
TIBURON(config-if)#
TIBURON(config-if)#exit
TIBURON(config)#interface Serial0/1/1
TIBURON(config-if)#ip address 172.18.110.198 255.255.255.252
TIBURON(config-if)#ip address 172.18.110.198 255.255.255.252
TIBURON(config-if)#no shutdown
TIBURON(config-if)#
TIBURON(config-if)#exit
TIBURON(config)#interface Serial0/1/0
TIBURON(config-if)#no shutdown
TIBURON(config-if)#
%LINK-5-CHANGED: Interface Serial0/1/0, changed state to up
TIBURON(config-if)#exit
TIBURON(config)#interface GigabitEthernet0/0
TIBURON(config-if)#no shutdown
TIBURON(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/0, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state to up
Copy
Paste
Top




```
Press RETURN to get started!



JERUNG>enable
JERUNG#
JERUNG#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
JERUNG(config)#interface GigabitEthernet0/0
JERUNG(config-if)#ip address 172.18.110.129 255.255.0.0
JERUNG(config-if)#ip address 172.18.110.129 255.255.255.224
JERUNG(config-if)#
JERUNG(config-if)#exit
JERUNG(config)#interface GigabitEthernet0/1
JERUNG(config-if)#ip address 172.18.110.161 255.255.255.224
JERUNG(config-if)#
JERUNG(config-if)#exit
JERUNG(config)#interface Serial0/0/0
JERUNG(config-if)#ip address 172.18.110.193 255.255.255.224
JERUNG(config-if)#ip address 172.18.110.193 255.255.255.252
JERUNG(config-if)#
JERUNG(config-if)#exit
JERUNG(config)#interface Serial0/0/1
JERUNG(config-if)#ip address 172.18.110.197 255.255.255.252
JERUNG(config-if)#ip address 172.18.110.197 255.255.255.252
JERUNG(config-if)#no shutdown
JERUNG(config-if)#
%LINK-5-CHANGED: Interface Serial0/0/1, changed state to up

JERUNG(config-if)#exit
JERUNG(config)#interface Serial0/0/0
JERUNG(config-if)#no shutdown
JERUNG(config-if)#
%LINK-5-CHANGED: Interface Serial0/0/0, changed state to up

JERUNG(config-if)#exit
JERUNG(config)#interface GigabitEthernet0/1
```

Task 2 – Routing Table

1. Paste the current routing table of each router here. There are 2 ways to this – via CLI and via Packet Tracer (PT) tool. Use both ways to show the routing table of all the routers. a. CLI

    i. Click on a router. In the prompt, type show ip route. The routing table of the router will be shown, as shown in Figure A.

```
SHARK#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

     172.18.0.0/16 is variably subnetted, 2 subnets, 2 masks
C       172.18.110.0/26 is directly connected, GigabitEthernet0/0
L       172.18.110.1/32 is directly connected, GigabitEthernet0/0
```

Figure A

```
SHARK#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

     172.18.0.0/16 is variably subnetted, 6 subnets, 3 masks
C       172.18.110.0/26 is directly connected, GigabitEthernet0/0
L       172.18.110.1/32 is directly connected, GigabitEthernet0/0
C       172.18.110.192/30 is directly connected, Serial0/0/0
L       172.18.110.194/32 is directly connected, Serial0/0/0
C       172.18.110.200/30 is directly connected, Serial0/0/1
L       172.18.110.201/32 is directly connected, Serial0/0/1
```

```
TIBURON#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

     172.18.0.0/16 is variably subnetted, 6 subnets, 3 masks
C       172.18.110.64/26 is directly connected, GigabitEthernet0/0
L       172.18.110.65/32 is directly connected, GigabitEthernet0/0
C       172.18.110.196/30 is directly connected, Serial0/1/1
L       172.18.110.198/32 is directly connected, Serial0/1/1
C       172.18.110.200/30 is directly connected, Serial0/1/0
L       172.18.110.202/32 is directly connected, Serial0/1/0
```

```
JERUNG#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

     172.18.0.0/16 is variably subnetted, 8 subnets, 3 masks
C       172.18.110.128/27 is directly connected, GigabitEthernet0/0
L       172.18.110.129/32 is directly connected, GigabitEthernet0/0
C       172.18.110.160/27 is directly connected, GigabitEthernet0/1
L       172.18.110.161/32 is directly connected, GigabitEthernet0/1
C       172.18.110.192/30 is directly connected, Serial0/0/0
L       172.18.110.193/32 is directly connected, Serial0/0/0
C       172.18.110.196/30 is directly connected, Serial0/0/1
L       172.18.110.197/32 is directly connected, Serial0/0/1
```

b. PT tool

i. Click on the magnifying glass (shown in Figure B), then click on a router, then choose Routing Table. A sample of the result is shown in Figure C.

Figure B

Routing Table for SHARK

Type	Network	Port	Next Hop IP	Metric
C	172.18.110.0/26	GigabitEthernet0/0	---	0/0
L	172.18.110.1/32	GigabitEthernet0/0	---	0/0

Figure C

Routing Table for SHARK

Type	Network	Port	Next Hop IP	Metric
C	172.18.110.0/26	GigabitEthernet0/0	---	0/0
L	172.18.110.1/32	GigabitEthernet0/0	---	0/0
C	172.18.110.192/30	Serial0/0/0	---	0/0
L	172.18.110.194/32	Serial0/0/0	---	0/0
C	172.18.110.200/30	Serial0/0/1	---	0/0
L	172.18.110.201/32	Serial0/0/1	---	0/0

Routing Table for TIBURON

Type	Network	Port	Next Hop IP	Metric
C	172.18.110.64/26	GigabitEthernet0/0	---	0/0
L	172.18.110.65/32	GigabitEthernet0/0	---	0/0
C	172.18.110.196/30	Serial0/1/1	---	0/0
L	172.18.110.198/32	Serial0/1/1	---	0/0
C	172.18.110.200/30	Serial0/1/0	---	0/0
L	172.18.110.202/32	Serial0/1/0	---	0/0

Routing Table for JERUNG

Type	Network	Port	Next Hop IP	Metric
C	172.18.110.128/27	GigabitEthernet0/0	---	0/0
L	172.18.110.129/32	GigabitEthernet0/0	---	0/0
C	172.18.110.160/27	GigabitEthernet0/1	---	0/0
L	172.18.110.161/32	GigabitEthernet0/1	---	0/0
C	172.18.110.192/30	Serial0/0/0	---	0/0
L	172.18.110.193/32	Serial0/0/0	---	0/0

2. Try to ping PC2 from PC1, paste the results here.

```
C:\>ping 172.18.110.190

Pinging 172.18.110.190 with 32 bytes of data:

Reply from 172.18.110.190: bytes=32 time<1ms TTL=127
Reply from 172.18.110.190: bytes=32 time<1ms TTL=127
Reply from 172.18.110.190: bytes=32 time=3ms TTL=127
Reply from 172.18.110.190: bytes=32 time<1ms TTL=127

Ping statistics for 172.18.110.190:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 3ms, Average = 0ms

C:\>
```

3. Try to ping PC4 from PC1, paste the results here.

```
C:\>ping 172.18.110.62

Pinging 172.18.110.62 with 32 bytes of data:

Reply from 172.18.110.129: Destination host unreachable.
Reply from 172.18.110.129: Destination host unreachable.
Reply from 172.18.110.129: Destination host unreachable.
Reply from 172.18.110.129: Destination host unreachable.

Ping statistics for 172.18.110.62:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),

C:\>
```

4. Explain the reason(s) behind the results.
   Question 1 it indicates successful packets forward since there is no packet loss. For question 2 it is showing destination host unreachable indicating that the devices are incorrectly configured.
5. What needs to be done to ensure all PCs can ping each other successfully?
   All routers must be up, the PC must have correct IP addresses, default gateway and subnet mask.

Task 2 – Routing Configuration

1. Let's start by opening the routing table (using the PT tool) for TIBURON and JERUNG. This is done to show changes to the routing table as configurations are made.

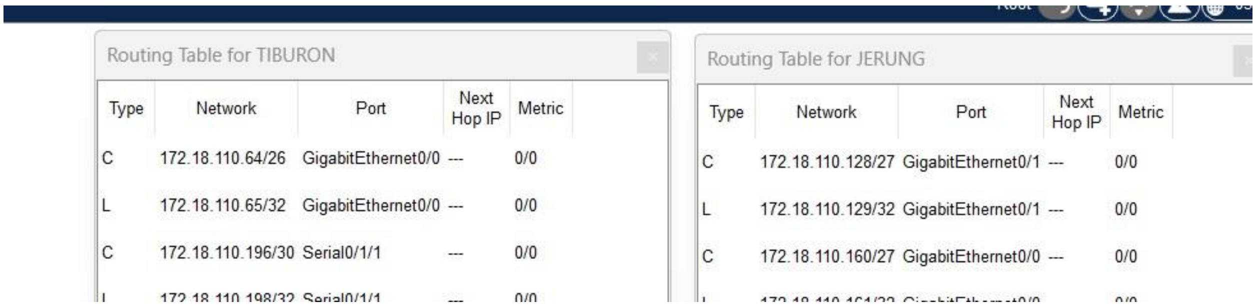


Figure D

2. In router JERUNG, configure the RIP routing protocol as shown in Figure E.

```
JERUNG#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
JERUNG(config)#router rip
JERUNG(config-router)#version 2
JERUNG(config-router)#network 172.18.110.128
JERUNG(config-router)#network 172.18.110.160
JERUNG(config-router)#network 172.18.110.192
JERUNG(config-router)#network 172.18.110.196
JERUNG(config-router)#no auto-summary
JERUNG(config-router)#
```

Figure E

   a. What can you say about the addresses used in the 'network' instructions in Figure E?
   The instruction is to direct to interconnected networks that JERUNG share with others router.

3. Then configure RIP in TIBURON. All are similar except use the network address. Use the instructions as shown below.

```
TIBURON (config-router) #network 172.18.110.64
TIBURON (config-router) #network 172.18.110.196
TIBURON (config-router) #network 172.18.110.200
```

4. As you may have seen, there are changes in the routing tables of both TIBURON and JERUNG. Paste a copy of these routing tables here.

Routing Table for JERUNG

Type	Network	Port	Next Hop IP	Metric
R	172.18.110.64/26	Serial0/0/1	172.18.110.198	120/1
C	172.18.110.128/27	GigabitEthernet0/0	---	0/0
L	172.18.110.129/32	GigabitEthernet0/0	---	0/0
C	172.18.110.160/27	GigabitEthernet0/1	---	0/0
L	172.18.110.161/32	GigabitEthernet0/1	---	0/0
C	172.18.110.192/30	Serial0/0/0	---	0/0
L	172.18.110.193/32	Serial0/0/0	---	0/0
C	172.18.110.196/30	Serial0/0/1	---	0/0
L	172.18.110.197/32	Serial0/0/1	---	0/0
R	172.18.110.200/30	Serial0/0/1	172.18.110.198	120/1

Routing Table for TIBURON

Type	Network	Port	Next Hop IP	Metric
C	172.18.110.64/26	GigabitEthernet0/0	---	0/0
L	172.18.110.65/32	GigabitEthernet0/0	---	0/0
R	172.18.110.128/27	Serial0/1/1	172.18.110.197	120/1
R	172.18.110.160/27	Serial0/1/1	172.18.110.197	120/1
R	172.18.110.192/30	Serial0/1/1	172.18.110.197	120/1
C	172.18.110.196/30	Serial0/1/1	---	0/0
L	172.18.110.198/32	Serial0/1/1	---	0/0
C	172.18.110.200/30	Serial0/1/0	---	0/0
L	172.18.110.202/32	Serial0/1/0	---	0/0

a. What are the changes seen in TIBURON?
3 routes to networks connected to JERUNG is discovered. The type R is the new route.

b. What are the Networks with type R in TIBURON and JERUNG?
In JERUNG it is 172.18.110.64/26, 172.18.110.200/30. For TIBURON it is 172.18.110.128/27, 172.18.110.160/27 and 172.18.110.192/30.

c. Ping PC3 from PC1. Was it successful?

```
C:\>ping 172.18.110.126

Pinging 172.18.110.126 with 32 bytes of data:

Reply from 172.18.110.126: bytes=32 time=9ms TTL=126
Reply from 172.18.110.126: bytes=32 time=2ms TTL=126
Reply from 172.18.110.126: bytes=32 time=5ms TTL=126
Reply from 172.18.110.126: bytes=32 time=3ms TTL=126

Ping statistics for 172.18.110.126:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 2ms, Maximum = 9ms, Average = 4ms
```

Successful.

d. Ping PC4 from PC1. Was it successful?

```
C:\>ping 172.18.110.62

Pinging 172.18.110.62 with 32 bytes of data:

Reply from 172.18.110.129: Destination host unreachable.
Request timed out.
Reply from 172.18.110.129: Destination host unreachable.
Reply from 172.18.110.129: Destination host unreachable.

Ping statistics for 172.18.110.62:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),
```

Not successful.

e. Explain the reasons for your answer.

Missing routes from PC1 to PC4.

f. Continue with configuration of RIP in SHARK. Paste your configurations here.

```
SHARK>enable
SHARK#
SHARK#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
SHARK(config)#router rip
SHARK(config-router)#version 2
SHARK(config-router)#network 172.18.110.0
SHARK(config-router)#network 172.18.110.192
SHARK(config-router)#network 172.18.110.200
SHARK(config-router)#no auto-summary
SHARK(config-router)#
```

g. Open router SHARK's routing table and paste here.

Routing Table for SHARK

Type	Network	Port	Next Hop IP	Metric
C	172.18.110.0/26	GigabitEthernet0/0	---	0/0
L	172.18.110.1/32	GigabitEthernet0/0	---	0/0
R	172.18.110.64/26	Serial0/0/1	172.18.110.202	120/1
R	172.18.110.128/27	Serial0/0/0	172.18.110.193	120/1
R	172.18.110.160/27	Serial0/0/0	172.18.110.193	120/1
C	172.18.110.192/30	Serial0/0/0	---	0/0
L	172.18.110.194/32	Serial0/0/0	---	0/0
R	172.18.110.196/30	Serial0/0/0	172.18.110.193	120/1
R	172.18.110.196/30	Serial0/0/1	172.18.110.202	120/1
C	172.18.110.200/30	Serial0/0/1	---	0/0
L	172.18.110.201/32	Serial0/0/1	---	0/0

h. Try to ping from PC4 to all other PCs in the topology. *Note: Try to ping at least twice to get best results.

```
C:\>ping 172.18.110.158

Pinging 172.18.110.158 with 32 bytes of data:

Reply from 172.18.110.158: bytes=32 time=7ms TTL=126
Reply from 172.18.110.158: bytes=32 time=6ms TTL=126
Reply from 172.18.110.158: bytes=32 time=4ms TTL=126
Reply from 172.18.110.158: bytes=32 time=6ms TTL=126

Ping statistics for 172.18.110.158:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 4ms, Maximum = 7ms, Average = 5ms

C:\>ping 172.18.110.190

Pinging 172.18.110.190 with 32 bytes of data:

Reply from 172.18.110.190: bytes=32 time=15ms TTL=126
Reply from 172.18.110.190: bytes=32 time=26ms TTL=126
Reply from 172.18.110.190: bytes=32 time=27ms TTL=126
Reply from 172.18.110.190: bytes=32 time=13ms TTL=126

Ping statistics for 172.18.110.190:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 13ms, Maximum = 27ms, Average = 20ms

C:\>ping 172.18.110.126

Pinging 172.18.110.126 with 32 bytes of data:

Reply from 172.18.110.126: bytes=32 time=27ms TTL=126
Reply from 172.18.110.126: bytes=32 time=28ms TTL=126
Reply from 172.18.110.126: bytes=32 time=25ms TTL=126
Reply from 172.18.110.126: bytes=32 time=55ms TTL=126

Ping statistics for 172.18.110.126:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 25ms, Maximum = 55ms, Average = 33ms
```

PC	Ping result
1	**Success**
2	**Success**
3	**Success**

Task 3 – Routing Update

1. Let's try a little experiment. Change the IP addresses of router TIBURON interface G0/0 to 192.168.1.1/24.
    a. This means that the subnet has changed. Find the new Network address of this subnet. Network Address is: **192.168.1.0/24**
    b. As this change happens, PC3 must also have different IP address, subnet mask and gateway address. What will it be?

PC3	Info
IP address	**192.168.1.2**
Subnet Mask	**255.255.255.0**
Gateway Address	**192.168.1.1/24**

    c. After this change, can PC4 and PC1 ping PC3?
    No, PC4 and PC1 cannot ping PC3.
    d. Copy and paste the routing tables for both SHARK and TIBURON here.

Routing Table for SHARK

Type	Network	Port	Next Hop IP	Metric
C	172.18.110.0/26	GigabitEthernet0/0	---	0/0
L	172.18.110.1/32	GigabitEthernet0/0	---	0/0
R	172.18.110.128/27	Serial0/0/0	172.18.110.193	120/1
R	172.18.110.160/27	Serial0/0/0	172.18.110.193	120/1
C	172.18.110.192/30	Serial0/0/0	---	0/0
L	172.18.110.194/32	Serial0/0/0	---	0/0
R	172.18.110.196/30	Serial0/0/0	172.18.110.193	120/1
R	172.18.110.196/30	Serial0/0/1	172.18.110.202	120/1
C	172.18.110.200/30	Serial0/0/1	---	0/0
L	172.18.110.201/32	Serial0/0/1	---	0/0

Routing Table for TIBURON

Type	Network	Port	Next Hop IP	Metric
R	172.18.110.0/26	Serial0/1/0	172.18.110.201	120/1
R	172.18.110.128/27	Serial0/1/1	172.18.110.197	120/1
R	172.18.110.160/27	Serial0/1/1	172.18.110.197	120/1
R	172.18.110.192/30	Serial0/1/1	172.18.110.197	120/1
R	172.18.110.192/30	Serial0/1/0	172.18.110.201	120/1
C	172.18.110.196/30	Serial0/1/1	---	0/0
L	172.18.110.198/32	Serial0/1/1	---	0/0
C	172.18.110.200/30	Serial0/1/0	---	0/0
L	172.18.110.202/32	Serial0/1/0	---	0/0
C	192.168.1.0/24	GigabitEthernet0/0	---	0/0
L	192.168.1.1/32	GigabitEthernet0/0	---	0/0

    e. Referring to the routing table, explain your findings.
    In TIBURON the new network is already added while in SHARK the new network has not been discovered yet.
    f. What is your next move to ensure end-to-end connectivity (i.e. all PCs can ping each other successfully)?
    Add the new network in RIP configuration.

g. Show your configurations in TIBURON to ensure end-to-end connectivity.

h. To ensure end-to-end connectivity, ping to all the PCs from PC3.

```
C:\>ping 172.18.110.158

Pinging 172.18.110.158 with 32 bytes of data:

Reply from 172.18.110.158: bytes=32 time=6ms TTL=126
Reply from 172.18.110.158: bytes=32 time=6ms TTL=126
Reply from 172.18.110.158: bytes=32 time=8ms TTL=126
Reply from 172.18.110.158: bytes=32 time=8ms TTL=126

Ping statistics for 172.18.110.158:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 6ms, Maximum = 8ms, Average = 7ms
```

```
C:\>ping 172.18.110.190

Pinging 172.18.110.190 with 32 bytes of data:

Reply from 172.18.110.190: bytes=32 time=4ms TTL=126
Reply from 172.18.110.190: bytes=32 time=5ms TTL=126
Reply from 172.18.110.190: bytes=32 time=6ms TTL=126
Reply from 172.18.110.190: bytes=32 time=6ms TTL=126

Ping statistics for 172.18.110.190:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 4ms, Maximum = 6ms, Average = 5ms

C:\>ping 172.18.110.62

Pinging 172.18.110.62 with 32 bytes of data:

Reply from 172.18.110.62: bytes=32 time=8ms TTL=126
Reply from 172.18.110.62: bytes=32 time=5ms TTL=126
Reply from 172.18.110.62: bytes=32 time=6ms TTL=126
Reply from 172.18.110.62: bytes=32 time=4ms TTL=126

Ping statistics for 172.18.110.62:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 4ms, Maximum = 8ms, Average = 5ms
```

PC	Ping result
1	**Success**
2	**Success**
4	**Success**

REFLECTION

What have you learned in this task?

From this task, I have learned that the during the IP configuration that the default gateway, IP address and subnet mask had to be correct or else PC cannot ping each other. I have also learned how to divide the subnet and determine which IP address to assign. Besides that, I also learned to do RIP to ensure every router know about every network.